Direction Sense - Questions & Detailed Solutions

Infosys Systems Engineer (SE) | 5-6 Nov 2026 Drive | 100% Pure SE Verified

Total Questions: 20 | **Total Marks:** 80

Section: Direction Sense

Problem 1 • A

PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

A
located to the West of B. C is located at North in between A and B. D is exactly to the South of B and also in line with B. In which direction of C is D located?

A North and South

C North and West

D South and West

Verified Answer: (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **(B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 2 • Point

PRACTICE VARIANT

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

Point
B is 7 km to the west of Point A. Point C is 12 km to the south of Point E. Point A is 5km to the east of Point C. Point D is 8 km to the north of Point C. Point G is 6 km to the west of Point D. Point H is 14 km to the south of Point G. If Siya is standing 6 km to the east of Point H and takes 1 hour to complete 9km distance then in what possible time she would reach Point E, if taken the shortest route?

A 35

B 60

C 180

Verified Answer: (D) **Explanation / Solution:** In such a case, the minimum distance between Siya and Point E is 18 km, so Slya will take 2 hours or 120 minutes to reach point E.

Problem 3 • Virendra's

PRACTICE VARIANT

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

Virendra's
house is 30 m to the left of the Shani Dev Temple, which is 40 m to the north of the Church. One day, Virendra takes the road which is parallel to the temple-Church road and walks for 50 m before turning to his right. He further walks for 20 m and reaches the Gym.
If
Virendra reaches the church taking the temple-church road and then turns 45 degrees thrice in the clock-wise direction, which direction would he be facing?

A Towards

C Towards

D North-East

✓ **Verified Answer:** (B) **Explanation / Solution:** From the final figure we can deduce the fact that Virendra must be facing his house.

Problem 4 • Read

PRACTICE VARIANT

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

Read
the given information carefully and answer the questions given beside.
P
+ Q states that P is 2 m East Of Q
P
^ Q states that P is 2 m South Of Q
P
& Q states that P is 4 m East Of Q
P
- Q states that P is 2 m West Of Q
P
/ Q states that P is 2 m North Of Q
A
– B ^ C & D + E, F + B
F
is in which direction with respect to E?

A North

B South

C Northwest

✓ **Verified Answer:** (D) **Explanation / Solution:** We can see from the figure that F is in South East of E.

Problem 5 • Some

PRACTICE VARIANT

ID: Q5 Source: Infosys PYQ Level: Infosys SE Standard

Some
chairs were placed in a park as per the manner described here. Chair B was 24m to the west of Chair A, which is 12m to the south of Chair E. Chair D was 10m to the east of Chair H. Chair C was 7m to the north of Chair H. Chair L was 15m to the north of Chair B, which is towards the south of Chair D. Chair F was 7m to the north of Chair E. Chair D and Chair E were placed in a straight horizontal line.
Chair
C is how much far (shortest) and in which direction from Chair F?

A 30m,

B 27m,

D 24m,

✓ **Verified Answer:** (C) **Explanation / Solution:** Chair C is 34m to the west of Chair F.

Problem 6 • Car

PRACTICE VARIANT

ID: Q6 Source: Infosys PYQ Level: Infosys SE Standard

Car
X starts from point A and travels in a straight line towards north for 8 km and takes a right turn and travels for 12 km then turns towards left and travels for 6 km before taking a right turn it travels for 10 km and takes a right turn and travels for 15 km and reaches a point B which is 8 km towards west of the point C where car Y stands. What is the position of Car Y with respect to initial position of car X?

A East

B Southwest

D South

✓ **Verified Answer:** (C) **Explanation / Solution:** Following the final solution, we can say that Car Y is southeast direction with respect to initial position of car X.

Problem 7 • Rudra

PRACTICE VARIANT

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

Rudra
started walking from his home in east direction and after walking for 80 m he took a turn of 135° towards his left and walked 22 m after that took a turn of 90° towards his right and walked 30 m and stopped. In which direction Ram is facing now?

A North

C Southwest

D East

Verified Answer: (B) **Explanation / Solution:** Here we can say that ram is facing in Northeast direction.

Problem 8 • Bonny

PRACTICE VARIANT

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

Bonny
moves 30 meters towards south then turns to his right and starts moving straight till he completes another 30 meters. Then turning to his left he moves for 20 meters. He again then turns to his left and moves for 30 meters. How far is he from his initial position?

A 30

C 60

D 10

Verified Answer: (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **(B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

Problem 9 • Point

PRACTICE VARIANT

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

Point
L is 2m towards north of point C. Point N is 7m towards east of point L. Point O is 8m towards south of point N. Point T is 12m towards west of point O. What is the direction of point C with respect to point T?

B Southwest

C North

D Southeast

Verified Answer: (A) **Explanation / Solution:** After observing the above figure, we can say that point C is in northeast direction with respect to point T.

Problem 10 • Box

PRACTICE VARIANT

ID: Q10 Source: Infosys PYQ Level: Infosys SE Standard

Box
B is in the north of Box C, which is in the north-east of Box D.
Box E is in the east of Box D. Box A is in the west of Box F,
which is in the northeast of Box B. Box E is in the middle of Box
F and Box G. Find Box A is in which direction from Box D?

A North

B Northeast

C South

Verified Answer: (D) **Explanation / Solution:** We cannot determine whether A is in north or north-east or north-west of D.

Problem 11 • The

PRACTICE VARIANT

ID: Q11 Source: Infosys PYQ Level: Infosys SE Standard

The
distance between N and P is 50 m and that between Q and R is 40m.
Q is to the west of R, which is to the east of N at a distance of
70m. P is to the west or northwest or north of Q. If Q is to the
south of P, what is the distance between P and Q?

B 35

C 80

D 20

Verified Answer: (A) **Explanation / Solution:** From Formula NP2 = NQ + PQ2 ∴ NP = NP2 – NQ2 = √ (50)2 – (30)2 = √2500 – 900 = √1600 = 40m

Problem 12 • The

PRACTICE VARIANT

ID: Q12 Source: Infosys PYQ Level: Infosys SE Standard

The
door of B's house faces east. From the back side of his
house, B walks straight 50 m. Again he walks 25m after turning to
his left, then he turns to his right and walks 50m. Finally he
turns towards north and walks 25m and stops.
Which
directions does B stand in at the end with respect to his house?

A North

B South

D East

Verified Answer: (C) **Explanation / Solution:** We can clearly observe from the figure that at the end point B stand in the west direction with respect to starting point.

Problem 13 • Some

PRACTICE VARIANT

ID: Q13 Source: Infosys PYQ Level: Infosys SE Standard

Some
chairs were placed in a park as per the manner described here. Chair B was 24m to the west of Chair A, which is 12m to the south of Chair E. Chair D was 10m to the east of Chair H. Chair C was 7m to the north of Chair H. Chair L was 15m to the north of Chair B, which is towards the south of Chair D. Chair F was 7m to the north of Chair E. Chair D and Chair E were placed in a straight horizontal line.
Find
the odd one out?

B B-D

C D-L

D H-C

✓ **Verified Answer:** (A) **Explanation / Solution:** F-B is the odd one out because the common thing among rest chairs is that the second chair is in the North of the first chair.

Problem 14 • Falyaz

PRACTICE VARIANT

ID: Q14 Source: Infosys PYQ Level: Infosys SE Standard

Falyaz
is standing to the north of Hiran. Hiran is standing 15m to the west of Mitul. Sahil is 15m to the north of Mitul and forms a straight horizontal line with Falyaz. Vihaan is equidistant from all of them. Vihaan is in which direction with respect to Mitul?

B North-East

C South-West

D South-East

✓ **Verified Answer:** (A) **Explanation / Solution:** Hiran is standing 15m to the west of Mitul. Sahil is 15m to the north of Mitul and forms a straight horizontal line with Falyaz. Falyaz is standing to the north of Hiran. Vihaan is equidistant from all of them. Note: As per statement, Vihaan is exactly in the mid-point of all 4 of them. To make exact midpoint, then the distance between Falyaz and Hiran should be 15m and also the distance between Falyaz and Sahil is 15m as shown in diagram. Vihaan is in North-West direction with respect to Mitul.

Problem 15 • Kundan

PRACTICE VARIANT

ID: Q15 Source: Infosys PYQ Level: Infosys SE Standard

Kundan
travels 15 km towards south and then takes a left turn and travels 10 km further again he takes a left turn and travels 15 km further. How far is he from his original position?

A 40km

B 25km

D 15km

✓ **Verified Answer:** (C) **Explanation / Solution:** ∵ AD = BC = 10 Km Therefore he is 10 km away from the original position.

Problem 16 • Shahrukh

PRACTICE VARIANT

ID: Q16 Source: Infosys PYQ Level: Infosys SE Standard

Shahrukh
starts driving from point A and drives 12 km towards north. He takes a right turn and drive 20 km. He now drives 8 km after taking a left turn. Finally he takes a left turn; and drives 20 km and stops at point B. How far is point A with respect to point B?

A 18

C 35

D 25

✓ **Verified Answer:** (B) **Explanation / Solution:** AB = AP + PB = 12 + 8 = 20 km

Problem 17 • Abhay

PRACTICE VARIANT

ID: Q17 Source: Infosys PYQ Level: Infosys SE Standard

Abhay
walks 20 m south from Point A to reach Point B. Then he turns to his right and walks 14 m to reach Point C. Again, he turns to his right and walks 20 m to reach Point D. Finally he turns to his left and walks 14 m to reach Point E. What is the distance between A and E and in which direction is Point E from Point A.

A 20

B 26

D 34

✓ **Verified Answer:** (C) **Explanation / Solution:** The distance between Point A and Point E = 14 + 14 = 28 m Following the image we can say that Point E is in west direction with respect to A

Problem 18 • Point

PRACTICE VARIANT

ID: Q18 Source: Infosys PYQ Level: Infosys SE Standard

Point
B is 25 m south of point A. Point C is 30 m east of point B. Point D is 15 m north of point C. Point E is 10 m west of point D.
If
point X is 30 m east of Point A, then Point D is how far and in which direction from point X?

A 10

B 5

C 15

Verified Answer: (D) **Explanation / Solution:** Following the common explanation we can say point D is 10 m away and towards south of point X.

Problem 19 • Lihaan

PRACTICE VARIANT

ID: Q19 Source: Infosys PYQ Level: Infosys SE Standard

Lihaan
walks 8m towards south from his office to reach park. Then take left and walks 15m to reach traffic signal. Then he turns 90 degree anti-clock wise direction and walks 8m to reach coffee shop. From the coffee shop he takes left and walks 8m to reach home. What is the shortest distance between Lihaan's office and Traffic signal?

A 30

B 25

D 15

Verified Answer: (C) **Explanation / Solution:** Shortest distance between Lihaan's office and Traffic signal = √82 + 152 = √64+225 = √289 =17m

Problem 20 • Point

PRACTICE VARIANT

ID: Q20 | Source: Infosys PYQ | Level: Infosys SE Standard

Point
A is 10m to the south of Point B, which is 8m to the west of Point C. Point D is 14m to the north of Point C. Point E is 4m to the east of Point F, which is 5m to the south of Point D. Point E is 8m to the west of Point G. Point H is 19m to the south of Point G. Point H is 12m to the east of Point I. Find the shortest distance between Points D and G?

B 13

C 12

D 20

✓ **Verified Answer:** (A) **Explanation / Solution:** The shortest distance between D and G is 13m.